

## Mark Scheme

Q1.

| Question                                                                                                                                                       | Scheme                                                          | Marks | AOs  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------|------|
| (a)                                                                                                                                                            | $2 < x < 6$                                                     | B1    | 1.1b |
|                                                                                                                                                                |                                                                 | (1)   |      |
| (b)                                                                                                                                                            | States either $k > 8$ or $k < 0$                                | M1    | 3.1a |
|                                                                                                                                                                | States e.g. $\{k : k > 8\} \cup \{k : k < 0\}$                  | A1    | 2.5  |
|                                                                                                                                                                |                                                                 | (2)   |      |
| (c)                                                                                                                                                            | Please see notes for alternatives                               |       |      |
|                                                                                                                                                                | States $y = ax(x-6)^2$ or $f(x) = ax(x-6)^2$                    | M1    | 1.1b |
|                                                                                                                                                                | Substitutes (2,8) into $y = ax(x-6)^2$ and attempts to find $a$ | dM1   | 3.1a |
|                                                                                                                                                                | $y = \frac{1}{4}x(x-6)^2$ or $f(x) = \frac{1}{4}x(x-6)^2$ o.e   | A1    | 2.1  |
|                                                                                                                                                                |                                                                 | (3)   |      |
| (6 marks)                                                                                                                                                      |                                                                 |       |      |
| <b>Notes:</b> Watch for answers written by the question. If they are beside the question and in the answer space, the one in the answer space takes precedence |                                                                 |       |      |



(a)

B1: Deduces  $2 < x < 6$  o.e. such as  $x > 2, x < 6$   $x > 2$  and  $x < 6$   $\{x : x > 2\} \cap \{x : x < 6\}$   $x \in (2, 6)$

Condone attempts in which set notation is incorrectly attempted but correct values can be seen or implied E.g.  $\{x > 2\} \cap \{x < 6\}$   $\{x > 2, x < 6\}$ . Allow just the open interval  $(2, 6)$

Do not allow for incorrect inequalities such as e.g.  $x > 2$  or  $x < 6$ ,  $\{x : x > 2\} \cup \{x : x < 6\}$   $x \in [2, 6]$

(b)

M1: Establishes a correct method by finding one of the (correct) inequalities

States either  $k > 8$  (condone  $k \geq 8$ ) or  $k < 0$  (condone  $k \leq 0$ )

Condone for this mark  $y \leftrightarrow k$  or  $f(x) \leftrightarrow k$  and  $8 < k < 0$

A1: Fully correct solution in the form  $\{k : k > 8\} \cup \{k : k < 0\}$  or  $\{k | k > 8\} \cup \{k | k < 0\}$  either way around

but condone  $\{k < 0\} \cup \{k > 8\}$ ,  $\{k : k < 0 \cup k > 8\}$ ,  $\{k < 0 \cup k > 8\}$ . It is not necessary to

mention  $\mathbb{R}$ , e.g.  $\{k : k \in \mathbb{R}, k > 8\} \cup \{k : k \in \mathbb{R}, k < 0\}$  Look for  $\{ \}$  and  $\cup$

Do not allow solutions not in set notation such as  $k < 0$  or  $k > 8$ .

(c)

M1: Realises that the equation of  $C$  is of the form  $y = ax(x-6)^2$ . Condone with  $a=1$  for this mark.

So award for sight of  $ax(x-6)^2$  even with  $a=1$

dM1: Substitutes  $(2, 8)$  into the form  $y = ax(x-6)^2$  and attempts to find the value for  $a$ .

It is dependent upon having an equation, which the  $(y = \dots)$  may be implied, of the correct form.

A1: Uses all of the information to form a correct equation for  $C$   $y = \frac{1}{4}x(x-6)^2$  o.e.

ISW after a correct answer. Condone  $f(x) = \frac{1}{4}x(x-6)^2$  but not  $C = \frac{1}{4}x(x-6)^2$ .

Allow this to be written down for all 3 marks

Examples of alternative methods

Alternative I part (c):

Using the form  $y = ax^3 + bx^2 + cx$  and setting up then solving simultaneous equations.

There are various versions of this but can be marked similarly

M1: Realises that the equation of  $C$  is of the form  $y = ax^3 + bx^2 + cx$  and forms two equations in  $a, b$  and  $c$ . Condone with  $a=1$  for this mark.

Note that the form  $y = ax^3 + bx^2 + cx + d$  is M0 until  $d$  is set equal to 0.

There are four equations that could be formed, only two are necessary for this mark.

Condone slips

Using  $(6, 0) \Rightarrow 216a + 36b + 6c = 0$

Using  $(2, 8) \Rightarrow 8a + 4b + 2c = 8$

Using  $\frac{dy}{dx} = 0$  at  $x = 2 \Rightarrow 12a + 4b + c = 0$

Using  $\frac{dy}{dx} = 0$  at  $x = 6 \Rightarrow 108a + 12b + c = 0$

dM1: Forms and solves three different equations, one of which must be using  $(2, 8)$  to find values for  $a, b$  and  $c$ . A calculator can be used to solve the equations

A1: Uses all of the information to form a correct equation for  $C$   $y = \frac{1}{4}x^3 - 3x^2 + 9x$  o.e.

ISW after a correct answer. Condone  $f(x) = \frac{1}{4}x^3 - 3x^2 + 9x$

**Alternative II part (c)**

**Using the gradient and integrating**

M1: Realises that the gradient of  $C$  is zero at 2 and 6 so sets  $f'(x) = k(x-2)(x-6)$  oe and attempts to integrate. Condone with  $k = 1$

dM1: Substitutes  $x = 2, y = 8$  into  $f(x) = k(\dots x^3 + \dots x + \dots)$  and finds a value for  $k$

A1: Uses all of the information to form a correct equation for  $C$   $y = \frac{3}{4}\left(\frac{1}{3}x^3 - 4x^2 + 12x\right)$  o.e.

ISW after a correct answer. Condone  $f(x) = \frac{3}{4}\left(\frac{1}{3}x^3 - 4x^2 + 12x\right)$

Q2.

| Question         | Scheme                      | Marks | AOs  |
|------------------|-----------------------------|-------|------|
| (a)              | $(-2, -3)$                  | B1    | 1.1b |
|                  |                             | (1)   |      |
| (b)              | $(-2, 5)$                   | B1    | 1.1b |
|                  |                             | (1)   |      |
| (c)              | Either $x = 0$ or $y = -13$ | M1    | 1.1b |
|                  | $(0, -13)$                  | A1    | 1.1b |
|                  |                             | (2)   |      |
| <b>(4 marks)</b> |                             |       |      |
| <b>Notes:</b>    |                             |       |      |

**Watch for answers in the body of the question and on sketch graphs. This is acceptable.**

**If coordinates are written by the question and in the main answer section the answer section takes precedence.**

(a)

B1: Accept without brackets. May be written  $x = -2, y = -3$

(b)

B1: Accept without brackets. May be written  $x = -2, y = 5$

(c)

M1: For either coordinate. E.g.  $(0, \dots)$  or  $(\dots, -13)$

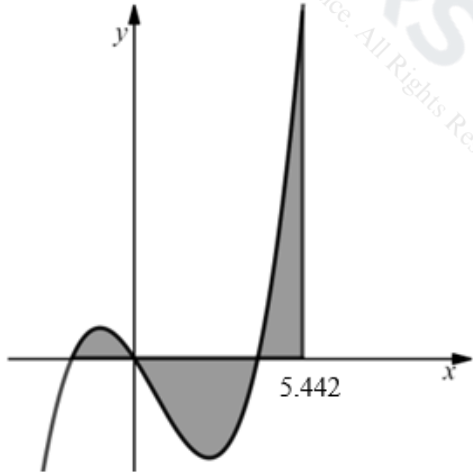
If they are building up their solution in stages e.g.  $(-2, -5) \rightarrow (0, -5) \rightarrow (0, -15) \rightarrow (0, -13)$   
only mark their final coordinate pair

A1: Correct coordinates. See above for building up solution in stages

Accept without brackets. May be written  $x = 0, y = -13$

SC 10 for candidates who write  $(-13, 0)$

Q3.

| Part | Working or answer an examiner might expect to see                                                           | Mark | Notes                                                                            |
|------|-------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------|
| (a)  | $y = x(x + 2)(x - 4) = x^3 - 2x^2 - 8x$                                                                     | B1   | This mark is given for expanding brackets as a first step to a solution          |
|      | $\int_{-2}^0 x^3 - 2x^2 - 8x \, dx$                                                                         | M1   | This mark is given for a method to find the exact area of $R_1$                  |
|      | $= \left[ \frac{1}{4}x^4 - \frac{2}{3}x^3 - 4x^2 \right]_{-2}^0$                                            | M1   | This mark is given for a method to evaluate the integral                         |
|      | $= 0 - (4 - \frac{-16}{3} - 16) = \frac{20}{3}$                                                             | A1   | This mark is given for a full method to show the exact value of $R_1$            |
| (b)  | $\frac{1}{4}b^4 - \frac{2}{3}b^3 - 4b^2 = -\frac{20}{3}$                                                    | M1   | This mark is given for deducing the area of $R_2 = -\frac{20}{3}$                |
|      | $3b^4 - 8b^3 - 48b^2 + 80 = 0$                                                                              | A1   | This mark is given for rearranging the equation to a quartic                     |
|      | $(b + 2)^2(3b^2 - 20b + 20)$<br>$= (b^2 + 4b + 4)(3b^2 - 20b + 20)$                                         | M1   | This mark is given for expanding the equation given                              |
|      | $= 3b^4 - 8b^3 - 48b^2 + 80 = 0$<br>The two equations are the same, so verified                             | A1   | This mark is for showing, and stating, that the equations are the same           |
| (c)  |                          | B1   | This mark is given for a sketch of the curve with $b = 5.442$ shown              |
|      | Between $x = -2$ and $b = 5.442$ , the area above the $x$ -axis is the same as the area below the $x$ -axis | B1   | This mark is given for a valid explanation of the significance of the root 5.442 |

Q4.

| Question                                                                                                                                                                                            | Scheme                                                                                                           | Marks | AOs  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)                                                                                                                                                                                                 | $f(x) \geq 5$                                                                                                    | B1    | 1.1b |
|                                                                                                                                                                                                     |                                                                                                                  | (1)   |      |
| (b)                                                                                                                                                                                                 | Uses $-2(3-x) + 5 = \frac{1}{2}x + 30$                                                                           | M1    | 3.1a |
|                                                                                                                                                                                                     | Attempts to solve by multiplying out bracket, collect terms etc<br>$\frac{3}{2}x = 31$                           | M1    | 1.1b |
|                                                                                                                                                                                                     | $x = \frac{62}{3}$ only                                                                                          | A1    | 1.1b |
|                                                                                                                                                                                                     |                                                                                                                  | (3)   |      |
| (c)                                                                                                                                                                                                 | Makes the connection that there must be two intersections.<br>Implied by either end point $k > 5$ or $k \leq 11$ | M1    | 2.2a |
|                                                                                                                                                                                                     | $\{k : k \in \mathbb{R}, 5 < k \leq 11\}$                                                                        | A1    | 2.5  |
|                                                                                                                                                                                                     |                                                                                                                  | (2)   |      |
| <b>(6 marks)</b>                                                                                                                                                                                    |                                                                                                                  |       |      |
| <b>Notes:</b>                                                                                                                                                                                       |                                                                                                                  |       |      |
| (a)                                                                                                                                                                                                 |                                                                                                                  |       |      |
| B1: $f(x) \geq 5$ Also allow $f(x) \in [5, \infty)$                                                                                                                                                 |                                                                                                                  |       |      |
| (b)                                                                                                                                                                                                 |                                                                                                                  |       |      |
| M1: Deduces that the solution to $f(x) = \frac{1}{2}x + 30$ can be found by solving<br>$-2(3-x) + 5 = \frac{1}{2}x + 30$                                                                            |                                                                                                                  |       |      |
| M1: Correct method used to solve their equation. Multiplies out bracket/ collects like terms                                                                                                        |                                                                                                                  |       |      |
| A1: $x = \frac{62}{3}$ only. Do not allow 20.6                                                                                                                                                      |                                                                                                                  |       |      |
| (c)                                                                                                                                                                                                 |                                                                                                                  |       |      |
| M1: Deduces that two distinct roots occurs when $y = k$ intersects $y = f(x)$ in two places. This may be implied by the sight of either end point. Score for sight of either $k > 5$ or $k \leq 11$ |                                                                                                                  |       |      |
| A1: Correct solution only $\{k : k \in \mathbb{R}, 5 < k \leq 11\}$                                                                                                                                 |                                                                                                                  |       |      |

Q5.

| Question  | Scheme                                                                                                                                                     | Marks      | AOs          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|
| (a)       | Either attempts $\frac{3x-7}{x-2} = 7 \Rightarrow x = \dots$                                                                                               | M1         | 3.1a         |
|           | Or attempts $f^{-1}(x)$ and substitutes in $x = 7$                                                                                                         |            |              |
|           | $\frac{7}{4}$ oe                                                                                                                                           | A1         | 1.1b         |
|           |                                                                                                                                                            | (2)        |              |
| (b)       | Attempts $ff(x) = \frac{3 \times \left( \frac{3x-7}{x-2} \right) - 7}{\left( \frac{3x-7}{x-2} \right) - 2} = \frac{3 \times (3x-7) - 7(x-2)}{3x-7-2(x-2)}$ | M1,<br>dM1 | 1.1b<br>1.1b |
|           | $= \frac{2x-7}{x-3}$                                                                                                                                       | A1         | 2.1          |
|           |                                                                                                                                                            | (3)        |              |
| (5 marks) |                                                                                                                                                            |            |              |
| Notes:    |                                                                                                                                                            |            |              |

(a)

**M1:** For either attempting to solve  $\frac{3x-7}{x-2} = 7$ . Look for an attempt to multiply by the  $(x-2)$  leading to a value for  $x$ .

Or score for substituting in  $x = 7$  in  $f^{-1}(x)$ . FYI  $f^{-1}(x) = \frac{2x-7}{x-3}$

The method for finding  $f^{-1}(x)$  should be sound, but you can condone slips.

**A1:**  $\frac{7}{4}$

(b)

**M1:** For an attempt at fully substituting  $\frac{3x-7}{x-2}$  into  $f(x)$ . Condone slips but the expression must

have a correct form. E.g.  $\frac{3 \times \left( \frac{*-*}{*-*} \right) - a}{\left( \frac{*-*}{*-*} \right) - b}$  where  $a$  and  $b$  are positive constants.

**dM1:** Attempts to multiply **all** terms on the numerator and denominator by  $(x-2)$  to create a fraction  $\frac{P(x)}{Q(x)}$

where both  $P(x)$  and  $Q(x)$  are linear expressions. Condone  $\frac{P(x)}{Q(x)} \times \frac{x-2}{x-2}$

**A1:** Reaches  $\frac{2x-7}{x-3}$  via careful and accurate work. Implied by  $a = 2, b = -7$  following correct work.

Methods involving  $\frac{3x-7}{x-2} = a + \frac{b}{x-2}$  may be seen. The scheme can be applied in a similar way

FYI  $\frac{3x-7}{x-2} = 3 - \frac{1}{x-2}$



Q6.

| Question  | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks    | AOs         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|
| (a)       | $k = e^2$ or $x \neq e^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | B1       | 2.2a        |
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (1)      |             |
| (b)       | $g'(x) = \frac{(\ln x - 2) \times \frac{3}{x} - (3 \ln x - 7) \times \frac{1}{x}}{(\ln x - 2)^2} = \frac{1}{x(\ln x - 2)^2}$ <p style="text-align: center;">or</p> $g'(x) = \frac{d}{dx} \left( 3 - (\ln(x) - 2)^{-1} \right) = (\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ <p style="text-align: center;">or</p> $g'(x) = (\ln x - 2)^{-1} \times \frac{3}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ <p>As <math>x &gt; 0</math> (or <math>1/x &gt; 0</math>) AND <math>\ln x - 2</math> is squared so <math>g'(x) &gt; 0</math></p> | M1<br>A1 | 1.1b<br>2.1 |
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | A1cso    | 2.4         |
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (3)      |             |
| (c)       | <p>Attempts to solve either <math>3 \ln x - 7 \dots 0</math> or <math>\ln x - 2 \dots 0</math> or <math>3 \ln a - 7 \dots 0</math> or <math>\ln a - 2 \dots 0</math> where <math>\dots</math> is “=” or “&gt;” to reach a value for <math>x</math> or <math>a</math> but may be seen as an inequality e.g. <math>x &gt; \dots</math> or <math>a &gt; \dots</math></p>                                                                                                                                                                                                                             | M1       | 3.1a        |
|           | $0 < a < e^2, a > e^{\frac{7}{3}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | A1       | 2.2a        |
|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (2)      |             |
| (6 marks) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |             |

**Notes:**

(a)

B1: Deduces  $k = e^2$  or  $x \neq e^2$  Condone  $k = \text{awrt } 7.39$  or  $x \neq \text{awrt } 7.39$

(b)

M1: Attempts to differentiate via the quotient rule and with  $\ln x \rightarrow \frac{1}{x}$  so allow for:

$$\frac{d}{dx}(g(x)) = \frac{(\ln x - 2) \times \frac{\alpha}{x} - (3 \ln x - 7) \times \frac{\beta}{x}}{(\ln x - 2)^2}, \beta > 0$$

But a correct rule may be implied by their  $u, v, u', v'$  followed by applying  $\frac{vu' - uv'}{v^2}$  etc.

Alternatively attempts to write  $g(x) = \frac{3 \ln(x) - 7}{\ln(x) - 2} = 3 - (\ln(x) - 2)^{-1}$  and attempts the chain rule so allow for:

$$3 - (\ln(x) - 2)^{-1} \rightarrow (\ln(x) - 2)^{-2} \times \frac{\alpha}{x}$$

Alternatively writes  $g(x) = (3 \ln(x) - 7)(\ln(x) - 2)^{-1}$  and attempts the product rule so allow for:

$$g'(x) = (\ln x - 2)^{-1} \times \frac{\alpha}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{\beta}{x}$$

In general condone missing brackets for the M mark. E.g. if they quote  $u = 3 \ln x - 7$  and  $v = \ln x - 2$  and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have  $v$  rather than  $v^2$  in the denominator.

A1:  $\frac{1}{x(\ln x - 2)^2}$  Allow  $\frac{\frac{1}{x}}{(\ln x - 2)^2}$  i.e. we need to see the numerator simplified to  $1/x$

Note that some candidates establish the correct numerator and correct denominator independently and provided they obtain the correct expressions, this mark can be awarded.

But allow a correctly expanded denominator.

A1cso: States that as  $x > 0$  AND  $\ln x - 2$  is squared so  $g'(x) > 0$

(c)

M1: Attempts to solve either  $3 \ln x - 7 = 0$  or  $\ln x - 2 = 0$  or using inequalities e.g.  $3 \ln x - 7 > 0$

A1:  $0 < a < e^2, a > e^{\frac{7}{3}}$

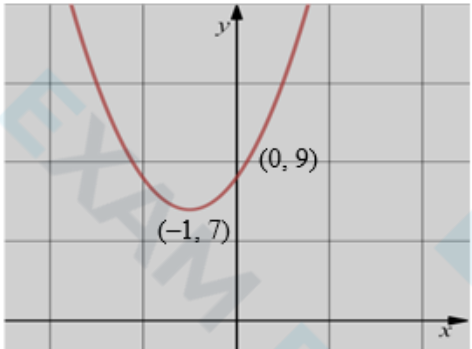
Q7.

| Part | Working or answer an examiner might expect to see                     | Mark | Notes                                                                                         |
|------|-----------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------|
| (a)  | The asymptote is found where $2x - q = 0$<br>Hence $q = 4$            | B1   | This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$ |
|      | $y = \frac{p-3x}{(2x-4)(x+3)}$ $\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$ | M1   | This mark is given for substituting $x = 3$ ,<br>$y = \frac{1}{2}$ (and $q = 4$ )             |

|     |                                                                                                                                                                             |    |                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------|
| (b) | $\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$                                                                                                            | M1 | This mark is given for a method to use partial fractions                                      |
|     | $= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$                                                                                                                                  | M1 | This mark is given for finding values for $A$ and $B$                                         |
|     | $= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$                                                                                                                                   | A1 | This mark is given for a fully simplified expression                                          |
|     | $I = \int \frac{15-3x}{(2x-4)(x+3)} dx$ $= m \ln(2x-4) + n \ln(x+3)$                                                                                                        | M1 | This mark is given for a method to integrate to find the area of $R$                          |
|     | $= 0.9 \ln(2x-4) + 2.4 \ln(x+3)$                                                                                                                                            | A1 | This mark is given for a correct expression for the area of $R$                               |
|     | $\text{Area } R = \left[ 0.9 \ln(2x-4) - 2.4 \ln(x+3) \right]_3^5$                                                                                                          | M1 | This mark is given for deducing an expression for the area of $R$<br>( $y = 0$ when $x = 5$ ) |
|     | $= [0.9 \ln 6 - 2.4 \ln 8] - [0.9 \ln 2 - 2.4 \ln 6]$ $= [0.9 \ln 6 + 2.4 \ln 6] - [7.2 \ln 2 + 0.9 \ln 2]$ $= 3.3 \ln 6 - 8.1 \ln 2$ $= 3.3 \ln 3 + 3.3 \ln 2 - 8.1 \ln 2$ | M1 | This mark is given for a method to find the exact area of $R$                                 |
|     | $= 3.3 \ln 3 - 4.8 \ln 2$                                                                                                                                                   | A1 | This mark is given for a correct value of the area of $R$ with $a = 3.3$ and $b = 4.8$        |



Q8.

| Part    | Working or answer an examiner might expect to see                                     | Mark | Notes                                                                                                       |
|---------|---------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------|
| (a)     | $2x^2 + 4x + 9 = 2(x + b)^2 + c$                                                      | B1   | This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$                             |
|         | $2x^2 + 4x + 9 = 2(x + 1)^2 + c$                                                      | M1   | This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$ and $b = 1$                 |
|         | $2x^2 + 4x + 9 = 2(x + 1)^2 + 7$                                                      | A1   | This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$ , $b = 1$ and $c = 7$       |
| (b)     |      | B1   | This mark is given for a U shaped curve in any position                                                     |
|         |                                                                                       | B1   | This mark is given for a y-intercept shown at (0, 9)                                                        |
|         |                                                                                       | B1   | This mark is given for a minimum shown at (-1, 7)                                                           |
| (c)(i)  | $g(x) = 2(x - 2)^2 + 4(x - 2) + 5$                                                    | M1   | This mark is given for writing $g(x)$ in the form $a(x + b)^2 + c$ and comparing to $f(x)$                  |
|         | Translation of $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$                                | A1   | This mark is given for deducing the translation of $y = f(x)$ to $y = g(x)$                                 |
| (c)(ii) | $h(x) = \frac{21}{2(x + 1)^2 + 7}$<br>Maximum value = $\frac{21}{7}$ (when $x = -1$ ) | M1   | This mark is given for writing $h(x)$ in the form $\frac{21}{a(x + b)^2 + c}$ and finding its maximum value |
|         | $0 < h(x) \leq 3$                                                                     | A1   | This mark is given for finding the correct range of the function $h(x)$                                     |

Q9.

| Question  | Scheme                                                                                                                                                                                                                                                                                                 | Marks    | AOs          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|
| (a)       | $f(x) = 4(x^2 - 2)e^{-2x}$                                                                                                                                                                                                                                                                             |          |              |
|           | Differentiates to $e^{-2x} \times 8x + 4(x^2 - 2) \times -2e^{-2x}$                                                                                                                                                                                                                                    | M1<br>A1 | 1.1b<br>1.1b |
|           | $f'(x) = 8e^{-2x} \{x - (x^2 - 2)\} = 8(2 + x - x^2)e^{-2x}$ *                                                                                                                                                                                                                                         | A1*      | 2.1          |
|           |                                                                                                                                                                                                                                                                                                        | (3)      |              |
| (b)       | States roots of $f'(x) = 0$ $x = -1, 2$                                                                                                                                                                                                                                                                | B1       | 1.1b         |
|           | Substitutes one $x$ value to find a $y$ value                                                                                                                                                                                                                                                          | M1       | 1.1b         |
|           | Stationary points are $(-1, -4e^2)$ and $(2, 8e^{-4})$                                                                                                                                                                                                                                                 | A1       | 1.1b         |
|           |                                                                                                                                                                                                                                                                                                        | (3)      |              |
| (c)       | (i) Range $[-8e^2, \infty)$ o.e. such as $g(x) \geq -8e^2$                                                                                                                                                                                                                                             | B1ft     | 2.5          |
|           | (ii) For <ul style="list-style-type: none"> <li>Either attempting to find <math>2f(0) - 3 = 2 \times -8 - 3 = (-19)</math> and identifying this as the <b>lower bound</b></li> <li>Or attempting to find <math>2 \times "8e^{-4}" - 3</math> and identifying this as the <b>upper bound</b></li> </ul> | M1       | 3.1a         |
|           | Range $[-19, 16e^{-4} - 3]$                                                                                                                                                                                                                                                                            | A1       | 1.1b         |
|           |                                                                                                                                                                                                                                                                                                        | (3)      |              |
| (9 marks) |                                                                                                                                                                                                                                                                                                        |          |              |
| Notes:    |                                                                                                                                                                                                                                                                                                        |          |              |

- (a)
- M1:** Attempts the product rule and uses  $e^{-2x} \rightarrow ke^{-2x}$ ,  $k \neq 0$
- If candidate states  $u = 4(x^2 - 2)$ ,  $v = e^{-2x}$  with  $u' = \dots$ ,  $v' = \dots e^{-2x}$  it can be implied by their  $vu' + uv'$
- If they just write down an answer without working award for  $f'(x) = pxe^{-2x} \pm q(x^2 - 2)e^{-2x}$
- They may multiply out first  $f(x) = 4x^2e^{-2x} - 8e^{-2x}$ . Apply in the same way condoning slips
- Alternatively attempts the quotient rule on  $f(x) = \frac{u}{v} = \frac{4(x^2 - 2)}{e^{2x}}$  with  $v' = ke^{2x}$  and  $f'(x) = \frac{vu' - uv'}{v^2}$
- A1:** A correct  $f'(x)$  which may be unsimplified.
- Via the quotient rule you can award for  $f'(x) = \frac{8xe^{2x} - 8(x^2 - 2)e^{2x}}{e^{4x}}$  o.e.
- A1\*:** Proceeds correctly to given answer showing all necessary steps.
- The  $f'(x)$  or  $\frac{dy}{dx}$  must be present at some point in the solution
- This is a "show that" question and there must not be any errors. All bracketing must be correct.
- Allow a candidate to move from the **simplified** unfactorised answer of  $f'(x) = 8xe^{-2x} - 8(x^2 - 2)e^{-2x}$



to the given answer in one step.

Do not allow it from an **unsimplified**  $f'(x) = 4 \times 2xe^{-2x} + 4(x^2 - 2) \times -2e^{-2x}$

Allow the expression / bracketed expression to be written in a different order.

So, for example,  $8(x - x^2 + 2)e^{-2x}$  is OK

(b)

**B1:** States or implies  $x = -1, 2$  (as the roots of  $f'(x) = 0$ )

**M1:** Substitutes one  $x$  value of their solution to  $f'(x) = 0$  in  $f(x)$  to find a  $y$  value.

Allow decimals here (3sf). FYI, to 3 sf,  $-4e^2 = -29.6$  and  $8e^{-4} = 0.147$

Some candidates just write down the  $x$  coordinates but then go on in part (c) to find the ranges using the  $y$  coordinates. Allow this mark to be scored from work in part (c)

**A1:** Obtains  $(-1, -4e^2)$  and  $(2, 8e^{-4})$  as the stationary points. This must be scored in (b). Remember to isw

after a correct answer. Allow these to be written separately. E.g.  $x = -1, y = -4e^2$

Extra solutions, e.g. from  $x = 0$  will be penalised on this mark.

(c)(i)

**B1ft:** For a correct range written using correct notation.

Follow through on  $2 \times$  their minimum "y" value from part (b), providing it is negative.

Condone a decimal answer if this is consistent with their answer in (b) to 3sf or better.

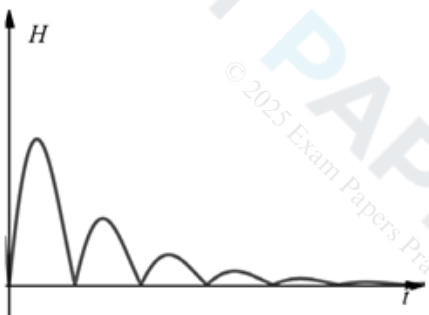
Examples of correct responses are  $[-8e^2, \infty)$ ,  $g \geq -8e^2$ ,  $y \geq -8e^2$ ,  $\{q \in \mathbb{R}, q \geq -8e^2\}$

(c)(ii)

**M1:** See main scheme. Follow through on  $2 \times$  their " $8e^{-4}$ "  $-3$  for the upper bound.

**A1:** Range  $[-19, 16e^{-4} - 3]$  o.e. such as  $-19 \leq y \leq 16e^{-4} - 3$  but must be exact

Q10.

| Part | Working or answer an examiner might expect to see                                                                    | Mark | Notes                                                                                                             |
|------|----------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------|
| (a)  | $f'(x) = -2.5e^{-0.25x} \sin x + 10e^{-0.25x} \cos x$                                                                | M1   | This mark is given for a method to differentiate to find an expression for $f'(x)$                                |
|      |                                                                                                                      | A1   | This mark is given for correctly differentiating to find an expression for $f'(x)$                                |
|      | $f'(x) = 0$<br>$\Rightarrow e^{-0.25x} (-2.5 \sin x + 10 \cos x) = 0$<br>$\Rightarrow (-2.5 \sin x + 10 \cos x) = 0$ | M1   | This mark is given for setting $f'(x) = 0$ and finding a method to solve for $\tan x$                             |
|      | $\frac{\sin x}{\cos x} = \frac{10}{2.5}$<br>$\tan x = 4$                                                             | A1   | This mark is given for showing that $\tan x = 4$ as required.                                                     |
| (b)  |                                    | M1   | This mark is given for a graph with a correct shape                                                               |
|      |                                                                                                                      | A1   | This mark is given for a graph with heights $> 0$                                                                 |
| (c)  | $\tan x = 4, x = 1.326$<br>$t = \pi + 1.326 = 4.47$                                                                  | M1   | This method is given for finding a value for $t$ between the first and second bounce                              |
|      | $H(4.47) =  10e^{-0.25 \times 4.47} \sin 4.47 $                                                                      | M1   | This mark is given for substituting the value of $t = \pi + \arctan 4$ into $H(t)$                                |
|      | $=  3.27 \times -0.97 $<br>$= 3.17 \text{ metres}$                                                                   | A1   | This mark is given for finding the maximum height of the ball                                                     |
| (d)  | The time between each bounce should not stay the same when the heights of each bounce are getting smaller            | B1   | This mark is given for a valid explanation of why the model should not be used to predict the time of each bounce |